WEEK 3 · LESSON 17 OF 24 · HANDOUT

Transformers and Large Language Models

Study time: **about 50 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=wjZofJX0v4M — open the lesson online to watch inside the course.

Your goal in this lesson: Explain how large language models work — tokens, embeddings, attention and next-token prediction — and describe how they are trained in stages.

A machine that predicts the next word

A **large language model (LLM)** — the engine behind ChatGPT, Gemini and Claude — does one basic job: given some text, it **predicts the next piece of text**. Repeat that prediction again and again, and it writes sentences, emails, code and essays.

This sounds simple, but to predict well across billions of examples, the model must learn grammar, facts, styles of writing and patterns of reasoning.

Step 1: tokens

Text is first cut into **tokens** — whole words, parts of words, or punctuation. "Murakoze cyane" may become several tokens. Each token gets a number ID. Languages with less training data, including Kinyarwanda, are often cut into more, smaller tokens, which can make models slower and less accurate for them.

Step 2: embeddings

Each token is turned into an **embedding**: a long list of numbers that captures its meaning. Words with similar meanings end up with similar numbers. "Teacher" and "lecturer" sit close together; "teacher" and "banana" sit far apart.

Step 3: attention — the key idea

The word "bank" means different things in "the bank gave me a loan" and "we sat on the river bank". To understand a word, you must look at the **other words around it**.

Attention lets every token look at every other token in the text and decide which ones matter most for its meaning. In "The farmer sold the cow because **it** was old", attention helps the model link "it" to "cow".

The **transformer** (introduced in 2017) is built from many layers of attention plus small neural networks. Because attention looks at all tokens at once instead of one by one, transformers train efficiently on GPUs — which is why they could be scaled up to huge sizes.

Step 4: predict, pick, repeat

At the end, the model gives a probability for every possible next token. One is chosen (a setting called **temperature** controls how adventurous the choice is), added to the text, and the process repeats.

How an LLM is trained

Stage	What happens	Result
1. Pre-training	Predict the next token on a huge amount of text from books, websites and code (self-supervised)	A model that knows language and many facts, but is not yet a helpful assistant
2. Fine-tuning	Train on examples of good questions and answers written by people	Follows instructions
3. RLHF / preference tuning	People rank answers; the model is tuned towards preferred ones (Lesson 11)	More helpful, polite and safer

Important limits

- **Hallucinations** — the model predicts *likely* text, not checked truth, so it can state false things confidently.
- **Knowledge cutoff** — it only knows what was in its training data, unless connected to search or documents.
- **Context window** — it can only read a limited amount of text at once.
- **Bias** — it can repeat unfair patterns from its training text.
- **Weaker in low-resource languages** — less training data means lower quality.

Beyond text

The same transformer ideas now power image generators, speech recognition, translation and **multimodal** models that handle text, images and audio together.

Remember this: An LLM turns text into tokens and embeddings, uses attention to understand context, and predicts the next token again and again. It is pre-trained on huge text, then fine-tuned and tuned with human feedback — and it can still be confidently wrong.

Name: _____ Date: _____

Activity

Task 1

Ask an AI assistant the same question three times. Note how the answers differ, and explain why using the idea of next-token probabilities.

Task 2

Ask it a question in Kinyarwanda and the same question in English. Compare the quality and write two sentences on what you notice.

Task 3

Write the sentence "The student gave the book to the teacher because she needed it." Who is "she"? Explain how attention helps a model decide.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What basic job does a large language model do?
   A. Predicts the next piece of text, again and again
   B. Searches a database of answers
   C. Copies Wikipedia exactly
   D. Translates only

2. What is a token?

A. A payment coin
B. A type of GPU
C. A user account
D. A piece of text such as a word or part of a word

3. What does attention allow a model to do?

A. Store more files
B. Look at other words in the text to understand each word's meaning
C. Work without data
D. Pay for electricity

4. What happens during pre-training?

A. The model learns to predict the next token on a huge amount of text
B. The model is deployed to phones
C. People rank answers
D. The model is deleted

5. Why can an LLM hallucinate?

A. It never makes mistakes
B. It is connected to the internet
C. It predicts likely text rather than checking facts
D. It has too few tokens

6. Why might an LLM perform worse in Kinyarwanda than in English?

A. Kinyarwanda uses numbers
B. LLMs cannot read African languages at all
C. Kinyarwanda has no grammar
D. There is much less Kinyarwanda training data, and words are split into more tokens

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